OL Organic Letters

pubs.acs.org/OrgLett Letter

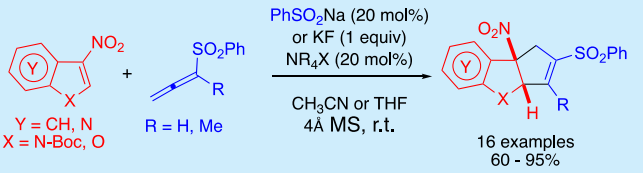
Dearomative (3 + 2) Cycloadditions of 3-Nitroheteroarenes with Allenyl Sulfones Mediated by Ion Pair Organocatalysis

Léo Birbaum, Moussa Ndiaye, Mahmoud Hachem, Stéphane Perrio, Michael De Paolis,* and Isabelle Chataigner*

Cite This: *Org. Lett.* 2025, 27, 1729–1734 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: We report the first example of dearomative (3 + 2) cycloadditions of 3-nitro(aza)-indole, -benzofuran, and -benzothiophene derivatives in the presence of allenyl sulfones, using sulfinate ammonium ion pairs as organocatalytic promoters. The methodology provides a new, facile, and efficient protocol for the synthesis of functionalized 2,3-fused cyclopentannulated indolines and dihydrobenzofurans.



Downloaded via ZHEJIANG UNIV on January 30, 2026 at 03:12:55 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

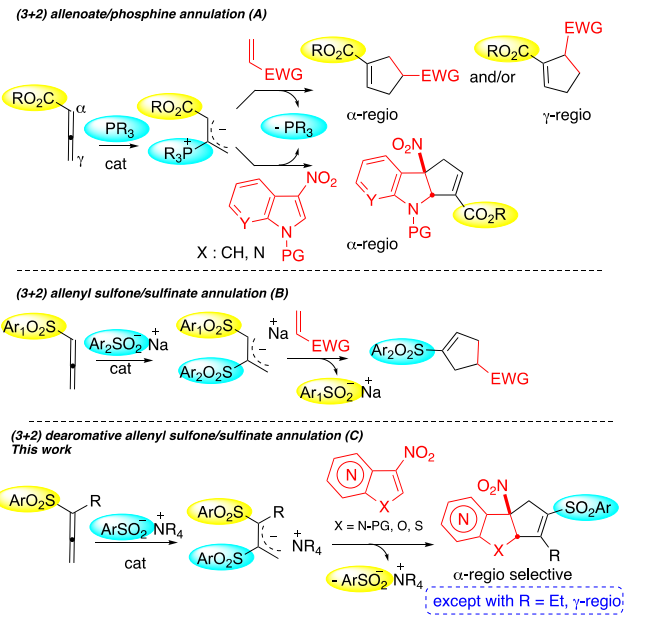
Dearomatization reactions are valuable tools in organic synthesis, generating three-dimensional (3D) molecules from easily accessible raw substrates.[1] However, the disruption of the stable aromatic π system makes their development challenging from both kinetic and thermodynamic viewpoints. In this context, electron-rich indoles have been used in many alkaloid syntheses, taking advantage of their C3 nucleophilicity.[2] In a complementary way, N-protected 3-nitroindoles have recently become privileged scaffolds, with their C2═C3 double bond displaying an electrophilic behavior akin to nitrostyrenes.[3] In the presence of electron-rich 1,3-dipoles, several dearomatizing (3 + 2) cycloadditions have been reported with 3-nitroindoles, allowing elegant generation of tricyclic indolines.[3a,4]

The addition of catalytic phosphines to allenoates generates 1,3-dipoles, which can react with Michael acceptors to form α- and/or γ-regioisomeric cyclopentenes (Scheme 1A, top).[5] Recently, it has been shown that nitroheteroaromatics also react with these dipoles to generate cyclopentaindolines, with an exclusive α-regioselectivity (Scheme 1A, bottom).[6]

As alternative electron-deficient allenes, allenyl sulfones exhibit distinct reactivity in (3 + 2) annulations in the presence of a sulfinate promoter (Scheme 1B).[7,8] This sulfinate is incorporated into the cycloadduct, while a sulfinate anion derived from the starting allenyl sulfone is released. As a result, sulfinate can be used in catalytic amounts if it is similar to the sulfonyl group on allene. These (3 + 2) annulations have been extended to a few classical (hetero)dipolarophiles, including acrylonitrile, methyl ketones, cyclic carbonyl compounds, or imines.[9] Noteworthy, employing a sulfinate−ammonium ($ArSO_2^-$ R_4N^+) cooperative ion pair as a promoter has proven beneficial, expanding the scope of this chemistry to a broader range of substrates.[9e−h]

Building upon our previous studies of cycloadditions with electrophilic arenes,[10] we now report the first dearomatization of 3-nitroheteroaromatics using allenyl sulfones, catalyzed by sulfinate ammonium ion pairs, leading to novel dearomatized sulfonyl-substituted cyclopentannulated heterocycles (Scheme 1C).

Scheme 1. (3 + 2) Annulations Involving Electron-Deficient Allenes



Received: January 17, 2025
Revised: January 31, 2025
Accepted: February 4, 2025
Published: February 8, 2025

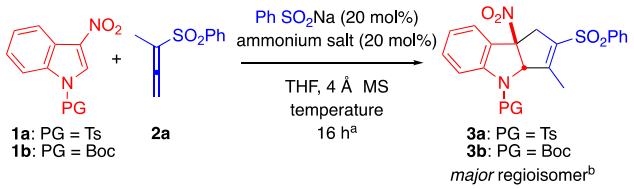


ACS Publications © 2025 American Chemical Society

https://doi.org/10.1021/acs.orglett.5c00215
Org. Lett. 2025, 27, 1729−1734

We began our study with 3-nitro-*N*-tosylindole **1a**, as a model of electrophilic arene,[3a] in the presence of α-methylallenyl sulfone **2a**, 20 mol % sodium benzenesulfinate ($PhSO_2Na$), and nBu_4NCl (entry 1 in Table 1). We were

Table 1. Optimization of the Reaction Conditions[a]



entry	indole	ammonium salt	temperature (°C)	conversion[b] (%)	yield (%)
1	**1a**	nBu_4NCl	45	35	
2	**1b**	nBu_4NCl	45	100	63
3	**1b**	nBu_4NCl	25	100	70
4	**1b**	nBu_4NCl	0	45	
5	**1b**	nBu_4NBr	25	60	50
6	**1b**[c]	nBu_4NCl	45	46	
7	**1b**[d]	$BnMe_3NCl$	25	90[e]	70
8	**1b**		45	0	
9	**1b**[f]	nBu_4NCl	45	0	

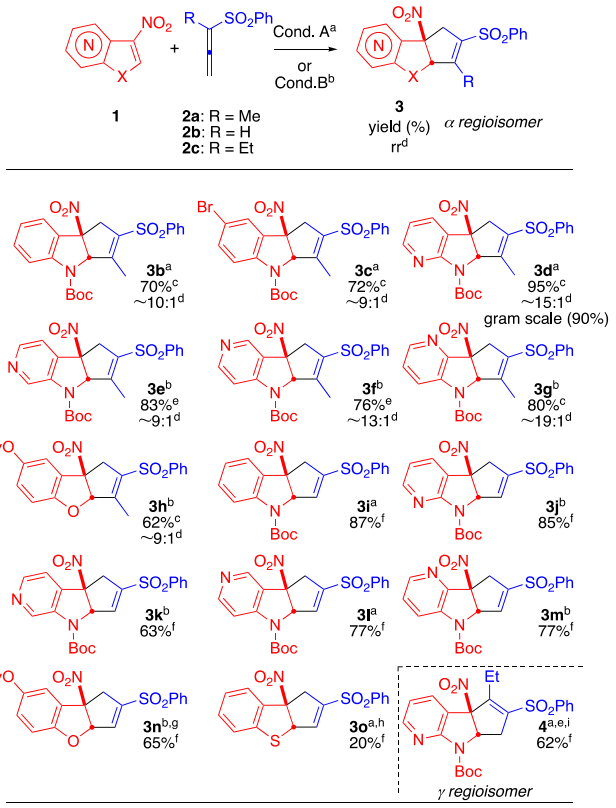
[a]Reaction conditions: compound **1** (1 equiv), compound **2a** (1.2 equiv), $PhSO_2Na$ (20 mol %), ammonium salt (20 mol %), and 4 Å molecular sieves (MS) in THF for 16 h. [b]At 9:1 rr. [c]With 10 mol % $PhSO_2Na$ and nBu_4NCl. [d]In CH_3CN for 12 h instead of THF for 16 h. [e]At 10:1 rr. [f]In the absence of $PhSO_2Na$.

pleased to observe the formation of compound **3a** at 45 °C, resulting from the cycloaddition of the transient allyl sulfone anion at the C2═C3 double bond of indole. Noteworthy, compound **3a** was stable under the reaction conditions, and no bis-adduct resulting from a subsequent addition to the newly formed electron-poor olefin could be detected. Optimization of the reaction conditions showed that *N*-Boc protection of indole was preferable (entry 2). Complete conversion of the starting material was observed, resulting in α- and γ-regioisomers [~9:1 regioisomeric ratio (rr)], with compound **3b** isolated in a 63% yield. Decreasing the temperature to room temperature led to a similar conversion but slightly higher yield without impact on the rr (entry 3). At a lower temperature, the reaction was less efficient (entry 4). Replacing the chloride counterion with bromine resulted in a lower conversion and yield (entry 5), while diminishing the catalyst loading was detrimental (entry 6). Notably, the nature of the quaternary ammonium salt can be easily changed, as the use of $BnMe_3NCl$ in CH_3CN for 12 h and nBu_4NCl in tetrahydrofuran (THF) for 16 h performed equally well (entry 7). Finally, the reaction required both the presence of sulfinate and ammonium catalysts (entries 8 and 9) (see the Supporting Information for additional details).

With the optimized conditions in hand, we then evaluated the scope of this new process with the variation of nitroarenes and allenyl sulfones (Scheme 2).

Reacting with compound **2a**, *N*-Boc indole **1b**, or the corresponding 5-bromoindole furnished the expected cycloadducts **3b** and **3c** in similar yields and α/γ-regioselectivities. 7-Aza-3-nitroindole **1c** led to compound **3d** in an enhanced 95% yield. A gram-scale experiment was performed with these reagents, yielding compound **3d** with a minimal loss of efficiency (90% yield). The isomers of the aza-nitro-indole

Scheme 2. (3 + 2) Annulations Involving 3-Nitro(aza)-indole, -benzofuran, and -benzothiophene Derivatives



[a]Reaction condition A: compound **1** (1 equiv), compound **2** (1.2 equiv), $PhSO_2Na$ (20 mol %), $BnMe_3NCl$ (20 mol %), and 4 Å MS in CH_3CN at room temperature or 0 °C for 16 h. [b]Reaction condition B: compound **1** (1 equiv), compound **2** (1.2 equiv), $PhSO_2Na$ (20 mol %), nBu_4NCl (20 mol %), and 4 Å MS in THF at room temperature for 16 h. [c]Isolated yield of the α-regioisomer. [d]rr was evaluated by ^1H nuclear magnetic resonance (NMR) on crude mixtures; see the Supporting Information for details. [e]Overall yield for the two regioisomers. [f]Only one regioisomer detected by ^1H NMR. [g]At 85% conversion. [h]At 50% conversion. [i]Formation of the γ-regioisomer.

scaffold were tolerated, and 6-, 5-, and 4-azaindoles were converted into the respective cycloadducts **3e**, **3f**, and **3g**, in 83, 76, and 80% yields, respectively. 3-Nitrobenzofuran was also converted into compound **3h** in 62% yield, exhibiting a 9:1 regioselectivity, similar to the results previously observed with (aza)nitroindoles.

Allenyl sulfone **2b** (R = H) was then tested in the cycloaddition process. A higher tendency to self-condensation was previously observed with this compound.[9b] Gratifyingly, under the optimized conditions, the reactivity of compound **2b** was selectively directed toward nitroheteroarenes, which were converted into cycloadducts **3i**−**3n** in good yields (63−87%) and exquisite α-regioselectivity in all cases. While inert toward α-methyl allenyl sulfone **2a**, 3-nitrobenzothiophene reacted with compound **2b** to give compound **3o** in 20% yield (50% conversion).

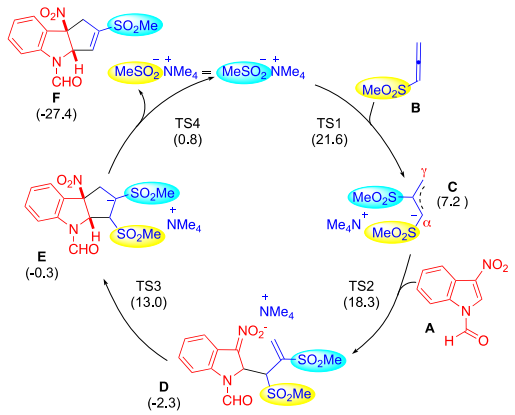
α-Ethylallenyl sulfone **2c** proved to be significantly less reactive than compounds **2a** and **2b**, showing no reaction with 3-nitroindole **1b**. In contrast, more electrophilic 7-aza-3-nitroindole **1c** was dearomatized at 0 °C in the presence of this more sterically hindered sulfone.[11] Strikingly, in this case,

https://doi.org/10.1021/acs.orglett.5c00215
Org. Lett. 2025, 27, 1729−1734

azaindoline **4**, isolated in 62% yield, displayed complete γ-regioselectivity, probably for steric reasons.

To have insight into the reaction mechanism, density functional theory (DFT) calculations were undertaken (see the Supporting Information for details). A model 3-nitroindole derivative, **A**, bearing a formyl group on the nitrogen center of the heterocycle, methanesulfonyl sulfone **B**, and a methanesulfinate anion associated with a tetramethylammonium cation were considered (Scheme 3).

Scheme 3. Proposed Mechanism and DFT Calculations[a]



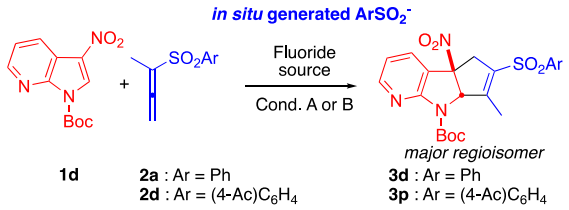
[a]Calculated at the (SMD: CH_3CN) M062X/6-31+G(d,p) level considering a polarized medium. Δ*G* values ($\Delta G^{\ddagger}$ and ΔG_r) in kcal mol^{-1} were computed as the difference between the separated reactants and the transition state (TS)/products.

The addition of methanesulfinate to allenyl sulfone **B** was found to be a highly activated associative step ($\Delta G^{\ddagger}$ = 21.6 kcal mol^{-1}), leading to allyl sulfone anion **C** (ΔG_r = 7.2 kcal mol^{-1}) in an endergonic process. The cycloaddition expectedly occurred in two steps, involving first the addition of carbanion at the C2 atom of indole **A** to generate nitronate **D** ($\Delta G^{\ddagger}$ = 18.3 kcal mol^{-1} and ΔG_r = −2.3 kcal mol^{-1}) and then its cyclization to give **E** ($\Delta G^{\ddagger}$ = 13.0 kcal mol^{-1} and ΔG_r = −0.3 kcal mol^{-1}). Then, the final step consisting of the sulfinate release was almost barrierless ($\Delta G^{\ddagger}$ = 0.8 kcal mol^{-1}) and highly exergonic (ΔG_r = −27.4 kcal mol^{-1}). While the formation of allyl sulfone anion **C** was more energetic than anticipated, the addition to indole appeared to be less energy-demanding than expected for a dearomatizing process, suggesting that it is a relatively facile step. Within the cycloaddition process, the first addition step (**A** + **C** → **D**) was the determining step, even if the cyclization step (**D** to **E**) exhibited a relatively high barrier ($\Delta G^{\ddagger}$ = 13.0 kcal mol^{-1}, with respect to separated substrates, and 15.3 kcal mol^{-1}, with respect to the adduct **D**), which was unexpected for an intramolecular cyclization. This is, however, in line with previous data computed with the related dearomatizing allenoate/phosphine (3 + 2) annulation process.[6f] The final step of the sequence appeared to be favorable from both kinetic and thermodynamic points of view. The stability of cycloadduct **F** probably drives the reaction to completion by shifting all possible equilibria and explains the whole transformation.

Considering the high energy barrier of the first step of the sequence, we then examined different nucleophilic promoters for the reaction. In this regard, sodium nitrites, cyanides, phosphines, or *N*-heterocyclic carbenes (NHCs) have already been employed to activate allenyl sulfones.[9] However, regardless of the promoter, only the formation of sulfonyl cycloadducts was observed, indicating that the acting catalyst in the process is the sulfinate anion itself, generated *in situ*.

To explore the reaction outcome with a different type of promoter, we considered fluorides to promote the addition of compound **1d** to compound **2a**. As a preliminary result, with nBu_4NF, no fluoro-containing product was formed beside the cycloadduct **3c**, isolated in 90% yield (entry 1 in Table 2) with

Table 2. (3 + 2) Annulations with an *In Situ* Generated Sulfinate Anion[a]



entry	allenyl sulfone	condition	conversion (%)	yield[b] (%)	α/γ ratio (3/4)
1	**2a**	A	100	90	∼7:3
2	**2a**	B	90	82	∼10:1
3	**2d**	B	80	68	∼10:1

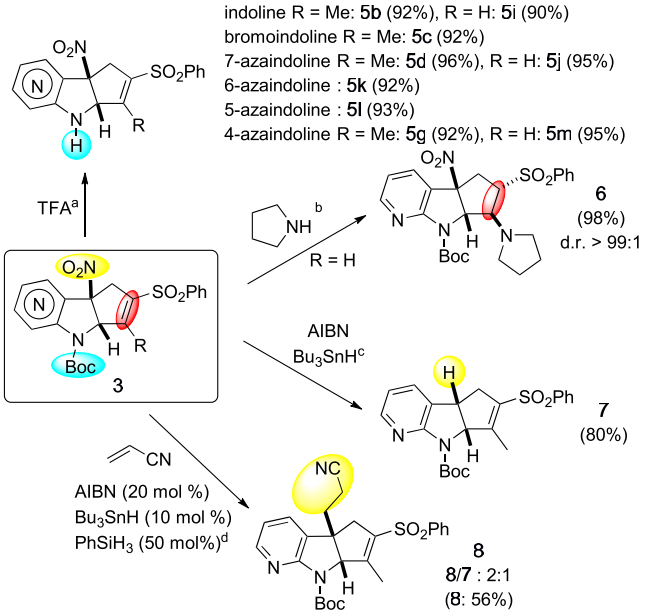
[a]Reaction condition A: compound **1** (1 equiv), compound **2** (1.5 equiv), TBAF (1 equiv), and 4 Å MS in THF at 0 °C for 12 h. Reaction condition B: compound **1** (1 equiv), compound **2** (1.5 equiv), KF (1 equiv), $BnMe_3NCl$, and 4 Å MS in CH_3CN at 0 °C for 16 h. [b]Overall yield.

the α-regioisomer remaining predominant (7:3 rr). This result suggests a mechanism involving the formation of a sulfinate anion triggered by the addition of fluoride ions to allenyl sulfone (see the Supporting Information for a proposed mechanism). To examine the impact of the quaternary ammonium salt, we repeated the reaction between compound **1d** and compound **2a** using KF and $BnMe_3NCl$ with success, as a similar yield in compound **3c** was obtained (entry 2). Equally striking was the higher rr measured, a value of 10:1 (α/γ) previously observed with this ammonium salt combined with $PhSO_2Na$. From a synthetic viewpoint, this fluoride-promoted method allows the formation of diverse arylsulfonylated cycloadducts without requiring the corresponding arenesulfinate anion, thus avoiding scrambling issues when using different aryl groups on allenyl sulfone and the sulfinate ion. In addition, it leads to volatile sub-products, which are easy to remove. To illustrate this point, allenyl sulfone **2d**, bearing a 4-acetylphenylsulfonyl group, was exposed to compound **1d** and the KF/$BnMe_3NCl$ combination. The corresponding cycloadduct **3p** was successfully isolated in 68% yield (entry 3).

Synthetic transformations were then performed along three distinct derivatizations (Scheme 4). First, the nitrogen atom protecting group was removed under acidic conditions, nicely affording (aza)indolines **5b**−**5m** in 90−96% yields. Second, we performed the aza-Michael addition of pyrrolidine, thereby demonstrating the electrophilic character of the vinylsulfone motif of compound **3i** and furnishing amino sulfone **6** in a quantitative yield and complete diastereoselectivity. As a last development, we sought to illustrate the versatility of the nitro group, which was pivotal for dearomatization. Hence, radical denitration (AIBN and nBu_3SnH) of compound **3d** led to

https://doi.org/10.1021/acs.orglett.5c00215
Org. Lett. 2025, 27, 1729−1734

Scheme 4. Post-functionalizations of (Aza)indolinic Cycloadducts 3



[a]Trifluoroacetic acid (TFA) and CH_2Cl_2 at room temperature for 4 h. [b]Pyrrolidine (5 equiv) and CH_2Cl_2 at room temperature for 12 h. [c]AIBN (0.8 equiv), nBu$_3$SnH (5 equiv), and toluene at 100 °C for 30 min. [d]AIBN (0.2 equiv), nBu$_3$SnH (0.1 equiv), $PhSiH_3$ (0.5 equiv), and 4 Å MS in benzene at 80 °C for 5 h, with 86% conversion.

azaindoline **7** in 80% yield. No rearomatization of the central five-membered ring was observed under these conditions. To further increase the molecular diversity, we devised to trap the hindered benzylic intermediate radical species with a Michael acceptor. The challenge of this transformation lies in the existing competition between alkylation and reduction with nBu$_3$SnH. After optimization, the best result was obtained when operating with a catalytic amount of nBu$_3$SnH (10 mol %) and $PhSiH_3$ (50 mol %). This method, described by Fu and co-workers,[12] was far only applied to the reductive denitration of nitroalkanes. Thus, in a rare example of intermolecular alkylative denitration,[13] compound **3d** reacted with acrylonitrile to give azaindoline **8** alongside denitrated compound **7** (**8**/**7** = 2:1) in 56% yield (86% conversion).

In conclusion, we have developed the first dearomatization of 3-nitroheteroarenes with allenyl sulfones to generate 2,3-fused cyclopentannulated heterocycles bearing a versatile vinylsulfone motif. The process is catalyzed by sulfinate anions, whether added or *in situ* generated, associated with a quaternary ammonium salt. Unsubstituted or α-methyl allenyl sulfones led to the exclusive or favored formation of the α-regioisomeric cycloadducts, while the less reactive more hindered α-ethyl allenyl sulfone furnished exclusively the γ-regioisomer in reaction with 7-aza-3-nitroindole. The mechanism, supported by a DFT study, suggests a highly energetic sulfonyl anion formation, followed by a two-step cycloaddition and a barrierless sulfinate release. The reactivity of azanitroindoles is especially interesting to exploit, revealing 3D molecules that are pharmaceutically relevant. The different derivatizations conducted further demonstrate the versatility of these cycloadducts. We are currently looking to extend the methodology to enantioselective processes through use of chiral ammonium salts.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00215.

Experimental procedures, compound characterization data, spectra, computational details, and Cartesian coordinates (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Michael De Paolis − *Université de Rouen Normandie, CNRS, Normandie Université, INSA Rouen Normandie, COBRA UMR 6014, INC3M FR 3038, F-76000 Rouen, France;* orcid.org/0000-0001-8139-3544; Email: michael.depaolis@univ-rouen.fr

Isabelle Chataigner − *Université de Rouen Normandie, CNRS, Normandie Université, INSA Rouen Normandie, COBRA UMR 6014, INC3M FR 3038, F-76000 Rouen, France; Sorbonne Université, CNRS, Laboratoire de Chimie Théorique, LCT UMR7616, 75005 Paris, France;* orcid.org/0000-0001-6302-7831; Email: isabelle.chataigner@univ-rouen.fr

Authors

Léo Birbaum − *Université de Rouen Normandie, CNRS, Normandie Université, INSA Rouen Normandie, COBRA UMR 6014, INC3M FR 3038, F-76000 Rouen, France*

Moussa Ndiaye − *Université de Rouen Normandie, CNRS, Normandie Université, INSA Rouen Normandie, COBRA UMR 6014, INC3M FR 3038, F-76000 Rouen, France*

Mahmoud Hachem − *Université de Rouen Normandie, CNRS, Normandie Université, INSA Rouen Normandie, COBRA UMR 6014, INC3M FR 3038, F-76000 Rouen, France*

Stéphane Perrio − *Université de Caen Normandie, ENSICAEN, CNRS, LCMT, Normandie Université, 14000 Caen, France;* orcid.org/0000-0002-3371-2405

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.5c00215

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Labex SynOrg (ANR-11-LABX-0029), Carnot Institute I2C, Graduate School for Research XL-Chem (ANR-18-EURE-0020 XL CHEM), Région Normandie Rouen University, CNRS, and INSA Rouen are thanked for financial support as well as CRIANN (Saint Etienne du Rouvray, France) for the generous allocation of computer time.

■ REFERENCES

(1) See for instance (a) Wertjes, W. C.; Southgate, E. H.; Sarlah, D. Recent Advances in Chemical Dearomatization of Nonactivated Arenes. *Chem. Soc. Rev.* **2018**, *47* (21), 7996−8017. (b) Zheng, C.; You, S.-L. Advances in Catalytic Asymmetric Dearomatization. *ACS Cent. Sci.* **2021**, *7* (3), 432−444. (c) Wang, J.; Luo, H.; Wang, X.; Wei, D.; Tian, R.; Duan, Z. Dearomatization [4 + 2] Cycloaddition of

https://doi.org/10.1021/acs.orglett.5c00215
Org. Lett. 2025, 27, 1729−1734

Nonactivated Benzene Derivatives. *Org. Lett.* **2022**, *24* (24), 4404−4408. (d) Escolano, M.; Gaviña, D.; Alzuet-Piña, G.; Díaz-Oltra, S.; Sánchez-Roselló, M.; Pozo, C. D. Recent Strategies in the Nucleophilic Dearomatization of Pyridines, Quinolines, and Isoquinolines. *Chem. Rev.* **2024**, *124* (3), 1122−1246.

(2) See for instance (a) Roche, S. P.; Youte Tendoung, J.-J.; Tréguier, B. Advances in Dearomatization Strategies of Indoles. *Tetrahedron* **2015**, *71* (22), 3549−3591. (b) Sheng, F.-T.; Wang, J.-Y.; Tan, W.; Zhang, Y.-C.; Shi, F. Progresses in Organocatalytic Asymmetric Dearomatization Reactions of Indole Derivatives. *Org. Chem. Front.* **2020**, *7* (23), 3967−3998. (c) Mei, G.-J.; Koay, W. L.; Tan, C. X. A.; Lu, Y. Catalytic Asymmetric Preparation of Pyrroloindolines: Strategies and Applications to Total Synthesis. *Chem. Soc. Rev.* **2021**, *50* (10), 5985−6012. (d) Wu, W.-T.; Ding, L.; Zhang, L.; You, S.-L. Gold-Catalyzed Intramolecular Dearomatization Reactions of Indoles for the Synthesis of Spiroindolenines and Spiroindolines. *Org. Lett.* **2020**, *22* (4), 1233−1238.

(3) For a review on 3-nitroindole, see for instance (a) Rkein, B.; Bigot, A.; Birbaum, L.; Manneveau, M.; De Paolis, M.; Legros, J.; Chataigner, I. Reactivity of 3-Nitroindoles with Electron-Rich Species. *Chem. Commun.* **2021**, *57* (1), 27−44. For the characterization of their electrophilic character, see (b) Rkein, B.; Manneveau, M.; Noël-Duchesneau, L.; Pasturaud, K.; Durandetti, M.; Legros, J.; Lakhdar, S.; Chataigner, I. How Electrophilic Are 3-Nitroindoles? Mechanistic Investigations and Application to a Reagentless (4 + 2) Cycloaddition. *Chem. Commun.* **2021**, *57* (78), 10071−10074. For recent examples of their reactivity in (4 + 2) dearomatization annulations, see for instance (c) Dou, P.-H.; Yuan, S.-P.; Chen, Y.; Zhao, J.-Q.; Wang, Z.-H.; You, Y.; Zhang, Y.-P.; Zhou, M.-Q.; Yuan, W.-C. Dearomatization of 3-Nitroindoles Enabled Using Palladium-Catalyzed Decarboxylative [4 + 2] Cycloaddition of 2-Alkylidenetrimethylene Carbonates. *J. Org. Chem.* **2022**, *87*, 6025−6037. (d) Zou, W.-L.; Liu, Y.-Q.; Li, Q.-Z.; Kou, X.-X.; Huang, H.; Tong, R.-S.; Li, J.-L.; Han, B. Dearomative Tandem Annulation to Access Chiral Indoline-Fused Bicyclo[2.2.2]Octanes Using Modularly Designed Organocatalysts. *Org. Chem. Front.* **2023**, *10*, 1237−1244. (e) Zhou, S.; Qian, H.-L.; Zhao, J.-Q.; You, Y.; Wang, Z.-H.; Yin, J.-Q.; Zhang, Y.-P.; Chen, M.-F.; Yuan, W.-C. Diastereoselective Synthesis of Polycyclic Indolines *via* Dearomative [4 + 2] Cycloaddition of 3-Nitroindoles with *Ortho* -Aminophenyl *p* -Quinone Methides. *Org. Biomol. Chem.* **2023**, *21*, 1373−1378. (f) Pandit, S.; Pandey, V. K.; Adhikari, A. S.; Kumar, S.; Maurya, A. K.; Kant, R.; Majumdar, N. Palladium-Catalyzed Dearomative [4 + 2]-Cycloaddition toward Hydrocarbazoles. *J. Org. Chem.* **2023**, *88* (1), 97−105. (g) Wang, K.; Li, Y.; Li, L.; Yao, W.; Li, Y.; Wang, W.; Chen, R.; Hu, Y.; Sun, A. Dearomative [4 + 2] Cycloaddition of 3-nitroindoles with *Ortho* -amino Morita−Baylis−Hillman Carbonates to Forge Indole-fused Quinolines. *J. Heterocyclic Chem.* **2024**, *61* (3), 528−537.

(4) See for instance (a) Wang, N.; Ren, J.; Li, K. Dearomatization of Nitro(hetero)arenes through Annulation. *Eur. J. Org. Chem.* **2022**, *2022* (18), No. e202200039. (b) Yuan, W.-C.; Chen, X.-M.; Zhao, J.-Q.; Zhang, Y.-P.; Wang, Z.-H.; You, Y. Ag-Catalyzed Asymmetric Interrupted Barton−Zard Reaction Enabling the Enantioselective Dearomatization of 2- and 3-Nitroindoles. *Org. Lett.* **2022**, *24* (3), 826−831. (c) Chen, Y.; Zhao, J.-Q.; Zhang, Y.-P.; Zhou, M.-Q.; Zhang, X.-M.; Yuan, W.-C. Copper-Catalyzed Asymmetric Dearomative [3 + 2] Cycloaddition of Nitroheteroarenes with Azomethines. *Molecules* **2023**, *28* (6), 2765. (d) Dou, P.-H.; Fu, X.-H.; Chen, Y.; Ge, Z.-Z.; Zhou, M.-Q.; Wang, Z.-H.; You, Y.; Yang, L.; Zhang, Y.-P.; Zhao, J.-Q.; Yuan, W.-C. Palladium-Catalyzed Asymmetric Decarboxylation of 5-Vinyloxazolidine-2,4-Diones Triggering the Dearomatization of Electron-Deficient Indoles for the Synthesis of Chiral Highly Functionalized Pyrroloindolines. *Org. Lett.* **2024**, *26* (15), 3310−3315.

(5) For seminal work, see (a) Zhang, C.; Lu, X. Phosphine-Catalyzed Cycloaddition of 2,3-Butadienoates or 2-Butynoates with Electron-Deficient Olefins. A Novel [3 + 2] Annulation Approach to Cyclopentenes. *J. Org. Chem.* **1995**, *60*, 2906−2908. For a review, see for instance (b) Wei, Y.; Shi, M. Lu's [3 + 2] Cycloaddition of Allenes with Electrophiles: Discovery, Development and Synthetic Application. *Org. Chem. Front.* **2017**, *4* (9), 1876−1890.

(6) (a) Wang, H.; Zhang, J.; Tu, Y.; Zhang, J. Phosphine-Catalyzed Enantioselective Dearomative [3 + 2]-Cycloaddition of 3-Nitroindoles and 2-Nitrobenzofurans. *Angew. Chem., Int. Ed.* **2019**, *58*, 5422−5426. (b) Li, K.; Gonçalves, T. P.; Huang, K.-W.; Lu, Y. Dearomatization of 3-Nitroindoles by a Phosphine-Catalyzed Enantioselective [3 + 2] Annulation Reaction. *Angew. Chem., Int. Ed.* **2019**, *58*, 5427−5431. (c) Jin, L.-W.; Jiang, F.; Chen, K.-W.; Du, B.-X.; Mei, G.-J.; Shi, F. Phosphine-Catalyzed Regiospecific (3 + 2) Cyclization of 3-Nitroindoles with Allene Esters. *Org. Biomol. Chem.* **2019**, *17*, 3894−3901. (d) Liu, K.; Wang, G.; Cheng, S.-J.; Jiang, W.-F.; He, C.; Ye, Z.-S. Phosphine-Catalyzed Dearomative [3 + 2] Annulation of 3-Nitroindoles and Allenoates. *Tetrahedron Lett.* **2019**, *60*, 1885−1890. (e) Cerveri, A.; Faza, O. N.; López, C. S.; Grilli, S.; Monari, M.; Bandini, M. Phosphine-Catalyzed Stereoselective Dearomatization of 3-NO_2-Indoles with Allenoates. *J. Org. Chem.* **2019**, *84*, 6347−6355. (f) Birbaum, L.; Gillard, L.; Gérard, H.; Oulyadi, H.; Vincent, G.; Moreau, X.; De Paolis, M.; Chataigner, I. Dearomatization of 3-Nitroindoles with Highly γ-Functionalized Allenoates in Formal (3 + 2) Cycloadditions. *Chem. - Eur. J.* **2019**, *25* (60), 13688−13693.

(7) For reviews, see for instance (a) Back, T. G. The Chemistry of Acetylenic and Allenic Sulfones. *Tetrahedron* **2001**, *57*, 5263−5301. (b) Yadav, D.; Menon, R. S. Recent Developments in the Chemistry of Allenyl Sulfones. *Org. Biomol. Chem.* **2020**, *18*, 365−378.

(8) For seminal work on these annulations, see (a) Padwa, A.; Yeske, P. E. Synthesis of Cyclopentenyl Sulfones via the [3 + 2] Cyclization-Elimination Reaction of (Phenylsulfonyl)allene. *J. Am. Chem. Soc.* **1988**, *110* (5), 1617−1618. (b) Padwa, A.; Yeske, P. E. [3 + 2] Cyclization-Elimination Route to Cyclopentenyl Sulfones Using (Phenylsulfonyl)-1,2-Propadiene. *J. Org. Chem.* **1991**, *56* (22), 6386−6390.

(9) See for instance (a) Moreno-Clavijo, E.; Carmona, A. T.; Reissig, H.-U.; Moreno-Vargas, A. J.; Alvarez, E.; Robina, I. Allenyl Sulfones and Allenyl Sulfides in the Synthesis of 3-Pyrrolines. A Novel Nucleophilic [3 + 2] Cycloaddition on Allenyl Sulfones Giving Rearranged Cycloadducts. *Org. Lett.* **2009**, *11* (21), 4778−4781. (b) Núñez, A.; Martín, M. R.; Fraile, A.; García Ruano, J. L. Abnormal Behaviour of Allenylsulfones under Lu's Reaction Conditions: Synthesis of Enantiopure Polyfunctionalised Cyclopentenes. *Chem. - Eur. J.* **2010**, *16* (18), 5443−5453. (c) Flasz, J. T.; Hale, K. J. A New Stereocontrolled Synthetic Route to (−)-Echinosporin from d -Glucose via Padwa Allenylsulfone [3 + 2]-Anionic Cycloadditive Elimination. *Org. Lett.* **2012**, *14* (12), 3024−3027. (d) Xiong, Z.; Hale, K. J. Total Synthesis of the Antitumor Macrolides, (+)-Brefeldin A and 4-Epi-Brefeldin A from d -Glucose: Use of the Padwa Anionic Allenylsulfone [3 + 2]-Cycloadditive Elimination to Construct Trans-Configured Chiral Cyclopentane Systems. *Org. Lett.* **2016**, *18* (17), 4254−4257. (e) Martzel, T.; Lohier, J.-F.; Gaumont, A.-C.; Brière, J.-F.; Perrio, S. Unique Reactivity of α-Substituted Electron-Deficient Allenes Using Sulfinate Salts as Lewis Base Organocatalysts. *Adv. Synth. Catal.* **2017**, *359* (1), 96−106. (f) Kuwano, S.; Masuda, T.; Yamaguchi, K.; Arai, T. N-Heterocyclic Carbene-Promoted [3 + 2] Cycloaddition of Allenyl Sulfone and Arylidenemalononitriles. *Heterocycles* **2017**, *95* (1), 232. (g) Martzel, T.; Lohier, J.; Gaumont, A.; Brière, J.; Perrio, S. Sulfinate-Organocatalyzed (3 + 2) Annulation Reaction of Propargyl or Allenyl Sulfones with Activated Imines. *Eur. J. Org. Chem.* **2018**, *2018*, 5069−5073. (h) Martzel, T.; Lohier, J.; Gaumont, A.; Brière, J.; Perrio, S. Sulfinate-Organocatalyzed (3 + 2) Annulation of Allenyl Sulfones with 1,1-Dicyano Olefins in the Presence of a Quaternary Ammonium Phase Transfer Agent. *Adv. Synth Catal* **2018**, *360* (14), 2696−2706.

(10) See for instance (a) Chataigner, I.; Hess, E.; Toupet, L.; Piettre, S. R. Activation of the Dienophilicity of Indoles in Normal Electron Demand [4 + 2] Cycloadditions under High Pressure. *Org. Lett.* **2001**, *3* (4), 515−518. (b) Chataigner, I.; Piettre, S. R. Multicomponent Domino [4 + 2]/[3 + 2] Cycloadditions of Nitroheteroaromatics: An Efficient Synthesis of Fused Nitrogenated Polycycles. *Org. Lett.* **2007**,

https://doi.org/10.1021/acs.orglett.5c00215
Org. Lett. 2025, 27, 1729−1734

9, 4159−4162. (c) Lee, S.; Chataigner, I.; Piettre, S. R. Facile Dearomatization of Nitrobenzene Derivatives and Other Nitroarenes with N-Benzyl Azomethine Ylide. *Angew. Chem., Int. Ed.* **2011**, *50* (2), 472−476. (d) Pasturaud, K.; Rkein, B.; Sanselme, M.; Sebban, M.; Lakhdar, S.; Durandetti, M.; Legros, J.; Chataigner, I. The Facile Dearomatization of Nitroaromatic Compounds Using Lithium Enolates of Unsaturated Ketones in Conjugate Additions and (4 + 2) Formal Cycloadditions. *Chem. Commun.* **2019**, *55* (52), 7494−7497. (e) Manneveau, M.; Tanii, S.; Gens, F.; Legros, J.; Chataigner, I. Dearomatization of 3-Cyanoindoles by (3 + 2) Cycloaddition: From Batch to Flow Chemistry. *Org. Biomol. Chem.* **2020**, *18* (18), 3481−3486. (f) Manneveau, M.; Rkein, B.; Cahard, D.; Legros, J.; Chataigner, I. Flow Dearomatization of Electron-Poor 3-Fluoromethylthioindoles by 1,3-Dipolar Cycloaddition. *J. Flow Chem.* **2022**, *12* (2), 141−145. (g) Rkein, B.; Coffinier, R.; Powderly, M.; Manneveau, M.; Sanselme, M.; Durandetti, M.; Sebban, M.; Hamdoun, G.; Oulyadi, H.; Harrowven, D.; Legros, J.; Chataigner, I. High Pressure Promoted Dearomatization of Nitroarenes by [4 + 2] Cycloadditions with Silyloxydienes. *Chem. Commun.* **2022**, *58* (84), 11807−11810. (h) Faye, Y.; Rkein, B.; Bigot, A.; Lequeux, T.; Legros, J.; Chataigner, I. Photocatalyzed (3 + 2) Cycloaddition for the Dearomatization of Electron-Poor Arenes under Flow Conditions. *Chem. - Eur. J.* **2023**, *29* (49), No. e202301567. (i) Powderly, M.; Roseau, M.; Frison, G.; Harrowven, D.; Legros, J.; Chataigner, I.; Hammami, R.; Chausset-Boissarie, L. Controlling Diastereoselectivity in Dearomatizing Diels-Alder Reactions of Nitroarenes with 2-Trimethylsilyloxycyclohexadiene. *Chem. - Eur. J.* **2024**, *30* (34), No. e202303697.

(11) At a higher temperature, the competing formation of 1,2-diphenylsulfonylpent-2-ene was predominant.

(12) Tormo, J.; Hays, D. S.; Fu, G. C. Bu_3SnH-Catalyzed Reduction of Nitroalkanes to Alkanes. *J. Org. Chem.* **1998**, *63* (16), 5296−5297.

(13) A recent contribution showed a photocatalytic denitrative alkenylation of nitroalkanes. See Osawa, A.; Uemura, K.; Murakami, S.; Nakao, Y. Photocatalytic Denitrative Alkenylation of Nitroalkanes. *Org. Lett.* **2024**, *26* (48), 10218−10223.

https://doi.org/10.1021/acs.orglett.5c00215
Org. Lett. 2025, 27, 1729−1734